1.1 IMPORTANCE OF CHEMISTRY

Chemistry plays a central role in science and is often intertwined with other branches of science. Principles of chemistry are applicable in diverse areas, such as weather patterns, functioning of brain and operation of a computer, chemical industries, manufacturing fertilisers, alkalis, acids, salts, dyes, polymers, drugs, soaps, detergents, metals, alloys and other inorganic and organic chemicals.

Maintaining the National Standards of Measurement

The system of units, including unit definitions, keeps on changing with time. Each modern industrialised country, including India, has a National Metrology Institute (NMI), which maintains standards of measurements. This responsibility has been given to the National Physical Laboratory (NPL), New Delhi.

1.2.1 STATES OF MATTER

Matter can exist in three physical states viz. solid, liquid and gas. The constituent particles of matter in these three states can be represented as shown below.

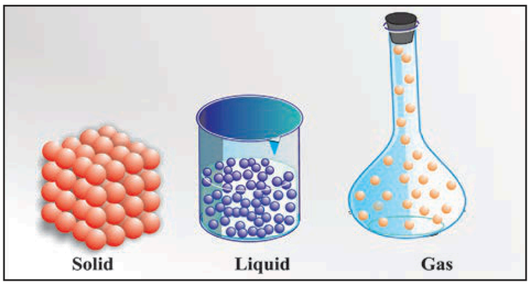

Fig. 1.1 Arrangement of particles in solid, liquid and gaseous state

Problem 1.1

A sample of drinking water was found to be severely contaminated with chloroform ($CHCl3$), supposed to be carcinogenic in nature. The level of contamination was 15 ppm (by mass). Express this in percent by mass.

1.3 DATA TABLES & MEASUREMENTS

Table 1.4 Data to Illustrate Precision and Accuracy

Measurements	1	2	Average (g)
Student A	1.95	1.93	1.940
Student B	1.94	2.05	1.995
Student C	2.01	1.99	2.000